

session_8hw_tsa

AUTHOR

Máté

```
library(fpp3)
```

— Attaching packages

fpp3 1.0.3 —

✓ tibble	3.3.1	✓ tsibble	1.2.0
✓ dplyr	1.2.1	✓ tsibbledata	0.4.1
✓ tidyr	1.3.2	✓ ggtime	1.0.0
✓ lubridate	1.9.5	✓ feasts	0.5.0
✓ ggplot2	4.0.3	✓ fable	0.5.0

— Conflicts —

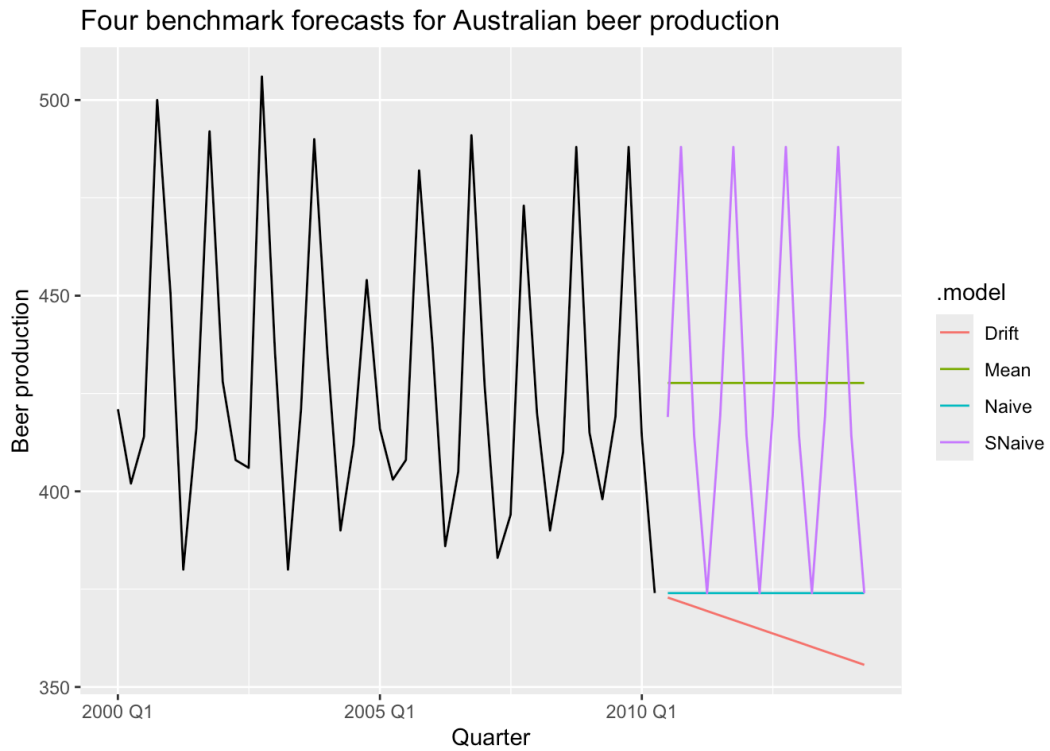
fpp3_conflicts —

* lubridate::date()	masks	base::date()
* dplyr::filter()	masks	stats::filter()
* tsibble::intersect()	masks	base::intersect()
* tsibble::interval()	masks	lubridate::interval()
* dplyr::lag()	masks	stats::lag()
* tsibble::setdiff()	masks	base::setdiff()
* tsibble::union()	masks	base::union()

```
beer <- aus_production |>
  filter(year(Quarter) >= 2000) |>
  select(Quarter, Beer)

fit <- beer |>
  model(
    Mean = MEAN(Beer),
    Naive = NAIVE(Beer),
    SNaive = SNAIVE(Beer),
    Drift = RW(Beer ~ drift())
  )

fit |>
  forecast(h = 16) |>
  autoplot(beer, level = NULL) +
  labs(
    title = "Four benchmark forecasts for Australian beer production",
    y = "Beer production"
  )
```



I would defend seasonal naive here, because there is a very strong seasonality in the data, so that seems to be the most likely accurate prediction. (I plotted 4 extra years (16 quarters) to see this even nicer, also I used data after 2000 but checked with starting from earlier and I made the same conclusion)